

BiasClean Phase 5 Validation

Six-Domain Real-Dataset Validation Report

AI Fairness CIO

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Pipeline version validated against: 3.10.10 (biasclean_v3_5_1_terminal.py)

1. Executive Summary

This report documents Phase 5 of BiasClean™'s external validation: running the pipeline against six real, publicly sourced datasets across the Finance, Health, Hiring, Education, Business, and Governance domains. Each dataset was validated twice, once in the native Python environment, once in the packaged Docker image (biasclean:3.10.5), to confirm the containerized deployment matches native behavior. Three domains (Finance, Health, Education) each contributed two datasets; the remaining three contributed one each, for nine dataset validations in total. Every dataset ran to completion; none were excluded (contrast with the Justice-domain report's opafy25nid.csv).

Domain	Dataset(s)	Native Result	Docker Result
Finance	HMDA (WA race cut)	0.0179 (unchanged)	0.0179 (unchanged), clean parity
Finance	German Credit	0.0429 (2 features)	0.0505→0.0158 (+69%), superseded by bucketing fix
Hiring	OpenIntro Resume	0.0608→0.0174 (+71%)	0.0608→0.0340 (+44%), SVM non-determinism, see §5
Health	Diabetes 130-US	0.0163 (unchanged)	0.0163 (unchanged), clean parity
Health	Heart Disease	0.0852→0.0301 (+65%)	0.0852→0.0301 (+65%), clean parity, Age resolved
Education	Bar Pass Prediction	0.0222 (below threshold)	0.0169, superseded by bucketing fix
Education	studentInfo (OULAD)	0.0458 (unchanged)	0.0816→0.0186 (+77%), clean parity, SVM correctly rejected (Region reversal)
Business	bank-full	0.0664→0.0776 (worse)	0.0169, superseded by bucketing fix
Governance	ACS Public Coverage	0.2036→0.1310 (+35.7%)	0.2036→0.1310 (+35.7%), clean parity

Six previously-latent pipeline bugs were found and fixed in the course of Phase 5 (v3.10.1 → v3.10.5): target-column auto-detection removal, delimiter auto-detection, camelCase/PascalCase tokenization, Finance SES keyword coverage plus p-value-based column selection, a rebalance-stage fuse preventing disadvantaged-group reversals, and continuous protected-attribute bucketing. The bucketing fix alone changed the validated outcome for three datasets (German Credit, Bar Pass, bank-full) after its introduction in v3.10.5, each confirmed on real data. A seventh finding, a native-macOS-vs-Docker result discrepancy in SVM training, was investigated during this validation round; full detail in Section 5. This finding was later reassessed in the v3.10.10 Docker sweep (see Section 4.7), the variance has not reproduced since Workstream D's dependency pinning.

The single most consequential finding across all six domains is Governance's rebalancing result: the pipeline's own fairness regulator excluded Ethnicity and Age from automatic correction after their rebalancing reversed the disadvantaged group, and separately rejected SVM enforcement entirely after it collapsed to a constant prediction. Two lower-severity, still-open transparency gaps in Governance's reporting, detailed in Section 6, remain unresolved as of the final Docker confirmation.

2. Methodology

All nine dataset validations were run against the BiasClean pipeline as it evolved from v3.10.1 through v3.10.5 over the course of Phase 5, with every dataset's final reported result reflecting v3.10.5 unless explicitly noted otherwise.

Non-technical-user policy, with one documented exception

As in the Justice-domain validation, interactive prompts were accepted at their default wherever possible. One dataset required a deliberate departure: German Credit specifically requires `auto_approve_threshold=0.70` (not the default 0.80) to correctly include its true SocioeconomicStatus columns at Tier 2 confidence. This was confirmed as the only Phase 5 dataset needing this override; all other datasets, native and Docker, ran at the default 0.80 threshold.

Two validation passes per dataset

Each dataset was run first in a native Python environment, then re-run inside the packaged Docker image (biasclean:3.10.5, confirmed built from a source file containing all six Phase 5 fixes) to verify containerized parity. Where the two passes disagree, this report states both numbers and the reason. In five of nine datasets, the passes matched exactly (clean parity). In three, Docker's result supersedes an earlier native run because the v3.10.5 bucketing fix postdated that native run and correctly fired in Docker. In one (Hiring), both numbers are legitimate but diverge due to a standing SVM non-determinism issue unrelated to the version difference, see Section 5.

Mode and SVM enforcement

Legacy mode (v2.7-equivalent rebalancing behavior) was used throughout for consistency. SVM fairness enforcement (fair_reduction method, wrapping SVC in fairlearn's ExponentiatedGradient) was applied wherever computationally feasible; the pipeline's own gate then independently accepts or rejects the SVM-stage result before it is allowed to become the delivered output.

Domain weighting

Each of the six domains carries its own Structural Inequality / Evidence Strength / Policy & Legal Relevance-derived weight table, per the same UK-specific framework documented in the Justice-domain report. Finance and Education weight SocioeconomicStatus highest (0.30 and 0.25 respectively); Governance weights Ethnicity highest (0.25); domain weightings are noted per-dataset below where they materially affect which feature drove a result.

Root-cause-first protocol

Unchanged from the Justice-domain validation: every anomaly was traced to its exact cause, via direct pipeline source inspection where needed, before being classified as either a dataset limitation, a Docker/native environment difference, or a pipeline defect requiring a fix and re-validation.

3. Per-Domain Results

3.1 Finance: HMDA (Washington State Race Cut)

Target column	loan_approved
Records	180,056 → 180,056 (100% retention; no rebalancing triggered)
Bias score (native, v3.10.1)	0.0179 (unchanged, below mitigation threshold)
Bias score (Docker, v3.10.5)	0.0179 (unchanged), clean parity, no Docker-specific discrepancies
SVM stage	Not applied, mitigation skipped

Feature	Column used	Weight	Significant
Ethnicity	derived_race	0.25	Yes
Gender	derived_sex	0.05	Yes

Finding: derived_race won Finance's single Ethnicity slot over derived_ethnicity. HMDA-style datasets need a separate Race vs. Hispanic-origin-Ethnicity category, which the pipeline's ontology does not currently provide (open item 3, unresolved).

Caveat: predicted_loan_approved was found tagged with a harmless, cosmetic __outcome__ mismatch that runs a pointless but inert protected-attribute validation (open item 6).

Finding: Composite stayed below the 0.05 mitigation threshold despite 2 statistically significant features correctly dual-gate behavior (per-feature significance and composite threshold evaluated separately), not a defect.

3.2 Finance: German Credit

Target column	CreditRisk
Records	1,000 → 1,000 (100% retention)
Configuration	auto_approve_threshold=0.70 (required override, the only Phase 5 dataset needing this)
Bias score (native, v3.10.4)	0.0429 (2 significant features: SES, Gender); below 0.05 threshold, no mitigation triggered
Bias score (Docker, v3.10.5)	0.0505 → 0.0158 (+69%); 3 significant features, SUPERSEDES the native record
SVM stage	Zero fuse exclusions needed
Rebalance gate	Accepted

Finding: Property was correctly selected for SocioeconomicStatus ($p=2.858e-05$) over SavingsAccountBonds despite the latter's lower raw p-value, correctly deferring to the small-group gate ($\text{min_group_size}=48 < 50$), confirms the full selection-tuple logic (small-group warning, then p-value, then confidence) working correctly in both native and Docker.

Finding: The Docker result differs from the native record because the v3.10.5 bucketing fix, introduced after the native run, also cleaned up this dataset's Age column (now 4 quantile buckets instead of raw per-year values), making Age reliably measurable and correctly newly flagged as significant. This is a bucketing-driven recalculation, not a regression.

Finding: Full mitigation succeeded in Docker: 0.0505→0.0158 (+69%). SES and Gender near-parity swaps were correctly left ungated; Age shows the one genuine trade-off regression (-22.9%), an expected compensatory pattern when other features improve.

Caveat: ForeignWorker/MigrationStatus keyword-coverage gap confirmed here ("foreign" not in the MigrationStatus pattern list), open item 4, not fixed.

This Docker result is now the current validated state for German Credit; the original v3.10.4 native result is retained for history only.

3.3 Hiring: OpenIntro Resume Callback Study

Target column	received_callback
Records	4,870 → 4,870 (100% retention)
Bias score (native, v3.10.1)	0.0608 → 0.0174 (+71%)
Bias score (Docker, v3.10.5)	0.0608 → 0.0340 (+44%)
SVM stage	fair_reduction, ACCEPTED by gate in both runs
Rebalance gate	Accepted (no actionable reversal; zero samples changed, groups already exactly balanced by construction)

Feature	Column used	Weight	p-value	Disparity: before → after
Ethnicity	race	0.25	4.76e-05	19.9pp → 0.35pp (native) / 0.24pp (Docker)
Gender	gender	0.20	0.382 (not sig.)	0.9pp → 7.4pp

Finding: Only 2 of this dataset's 33 columns map to a protected attribute (race, gender); the rest are resume-quality attributes, not protected-attribute candidates expected for this dataset's design (the Bertrand–Mullainathan resume/callback audit study). Ethnicity is significant and dramatically corrected (98%+ improvement in both runs); Gender is correctly not flagged, and report wording accurately distinguishes "unlikely due to chance" from "could plausibly be due to chance alone."

Finding: This is the first Phase 5 dataset (native or Docker) where SVM fairness enforcement actually passed its own validation gate rather than being correctly rejected for a classifier collapse, the first real-data confirmation that the gate can accept a genuinely good result, not only reject broken ones.

Finding, result reproducibility: The native and Docker runs, on byte-identical data with identical rebalancing and identical gate decisions, produced two different SVM-stage outcomes: 2,164 predictions changed (44.4%, native) versus 1,612 (33%, Docker). Root-caused by direct inspection of the pipeline source across both versions plus fairlearn's installed API: ExponentiatedGradient.fit() accepts no random-seed parameter at all, so while the pipeline seeds .predict(), the training step itself is inherently stochastic. Confirmed byte-for-byte identical (and identically unseeded) in both v3.10.1 and v3.10.5 source, this is a standing characteristic of the fair_reduction method, not a version regression or a Docker-specific defect. Both reported numbers are legitimate outcomes of the same non-deterministic training step, not a discrepancy between environments. Full detail in Section 5.

Caveat: Gender's apparent gap-widening after SVM (0.9pp→7.4pp, still non-significant, still non-reversed) is a symptom of the same SVM-training variability above, rather than an independent finding.

3.4 Health: Diabetes 130-US Hospitals

Target column	readmitted_30d
Records	99,492 → 99,492 (100% retention; no rebalancing triggered)
Bias score (native, v3.10.4)	0.0163 (unchanged)
Bias score (Docker, v3.10.5)	0.0163 (unchanged), clean parity, no Docker-specific discrepancies
SVM stage	Not applied, mitigation skipped

Feature	Weight	p-value	Significant
Ethnicity	0.25	0.1798 (native) / 0.180 (Docker)	No
Gender	0.05	0.3489 (native) / 0.349 (Docker)	No

Finding: Both features are genuinely non-significant in both runs, matching almost exactly. This confirms the pipeline's wording correctly branches to the true "nothing found" message when that is actually the case: a clean, no-bugs result.

3.5 Health: Heart Disease

Target column	heart_disease (binarized from the original 5-category 'target' column, coerce_binary_target correctly rejected the raw column first)
Records	303 → 303 (100% retention)
Bias score (native, v3.10.4)	0.0852 → 0.0301 (+65%)
Bias score (Docker, v3.10.5)	0.0852 → 0.0301 (+65%), clean parity
SVM stage	Applied, REJECTED (composite-score regression: 0.0301→0.0345)
Rebalance gate	Accepted, with fuse exclusion (Age)

Feature	Weight	p-value	Disparity: before → after
Gender	0.20	1.26e-06	55.3% vs 25.8% → 97.7% improvement
Age	0.15	1.86e-05	Bucketed into 4 ranges (29-48/48-56/56-61/61-77, min group size 70)

Finding: Age previously reported as "absent from mappings entirely" in an earlier, unconfirmed native attempt (the log initially sent was actually the diabetes run's log by mistake) is now clearly mapped and correctly flagged significant, resolving that ambiguity.

Finding: The rebalance-stage fuse engaged correctly: Age's rebalancing reversed the disadvantaged group ('29-48'→'56-61'), so it was excluded and reverted, leaving only Gender rebalanced. This is the third real-data cross-domain confirmation of the fuse (after Governance and German Credit).

Finding: SVM was applied and rejected for a third, distinct reason not seen elsewhere in Phase 5, not a classifier collapse, not a new reversal, but a plain composite-score regression (new_reversals: []). This is a good confirmation that the gate's multiple rejection criteria function independently rather than only catching collapse or reversal cases. Final predictions correctly fell back to the rebalanced stage.

3.6 Education: Bar Pass Prediction

Target column	pass_bar
Records	22,407 → 22,407 (100% retention)
Bias score (native, v3.10.4)	0.0222 (3 significant features; below 0.05 threshold, no mitigation triggered)
Bias score (Docker, v3.10.5)	0.0169, SUPERSEDES the native record
SVM stage	Not applied (composite below threshold in both runs)

Finding: Bucketing fired on both Age candidates in Docker: DOB_yr → 4 groups (min group size 603) and age → 4 groups (min group size 3,914). Both mapping-tie tie-breaks reproduced identically to the native record: Gender's gender(95%) vs sex(85%) resolved by confidence; Age's DOB_yr vs age tie (both 95% confidence) resolved by p-value (DOB_yr wins), consistent in both runs.

Finding: The composite difference (0.0169 Docker vs 0.0222 native) traces directly to DOB_yr's min_group_size jumping from 1 (native's most extreme instance of the raw-per-year open item) to 603 under bucketing, an expected bucketing-driven recalculation, not a bug. Both runs retain the same 3 significant features (Ethnicity/Gender/Age).

Caveat: The losing 'age' candidate bucketed into nonsensical negative ranges (-58 to -3, etc.). Since it lost the tie-break to DOB_yr, this doesn't affect the delivered result, but suggests this dataset's raw 'age' column may not be literal age-in-years (possibly a derived/normalized LSAC variable), worth a look sometime, not urgent.

3.7 Education: studentInfo (OULAD)

Target column	completed (binarized from final_result via .isin(['Pass','Distinction']))
Records	32,593 → 32,614 (100.1% retention)
Bias score (native, v3.10.4)	0.0816→0.0186 (+77%; 5 significant features)
Bias score (Docker, v3.10.5)	0.0816→0.0186 (+77%), matches native exactly; SocioeconomicStatus (imd_band) now included via the v3.10.6 keyword fix
SVM stage	Applied, REJECTED (composite worsened 0.0186→0.0440 plus new Region reversal); rebalanced stage used

Feature	p-value	Group rates
Gender	5.19e-05	F 48.4% vs M 46.2% (2.2pp gap, widened to 3.6pp after rebalancing, compensatory regression)
Region	4.71e-39	Ireland 54.9% vs North Western Region 40.2% (14.7pp gap, narrowed to 7.6pp after rebalancing)
Age	3.24e-34	55+: 61.6% vs 0-35: 45.0% (16.5pp gap, narrowed to 6.3pp after rebalancing), min_group_size=216
DisabilityStatus	4.15e-27	N: 48.2% vs Y: 38.1% (10pp gap, narrowed to 0.0pp after rebalancing, near-total resolution)
SocioeconomicStatus	7.17e-116	90-100%: 57.5% vs 0-10%: 35.2% (22.4pp gap, narrowed to 2.0pp after rebalancing — held as non-actionable, near-parity), min_group_size=2,536

Finding: This dataset ships age pre-bucketed (age_band, 3 categories), so the v3.10.5 bucketing fix correctly recognized it as already-fine and left it untouched in Docker, consistent, expected behavior, and a useful real-world illustration of exactly what the bucketing fix was built to replicate automatically elsewhere.

Finding: Age and DisabilityStatus are both genuinely and practically significant (dual_criteria_significant=true), representing real, meaningful disparities rather than large-sample statistical noise. Region and Gender are statistically significant but correctly stay below practical significance via the dual-criteria check.

Caveat: Resolved (v3.10.6): imd_band is now recognized by the Education ontology's SocioeconomicStatus keyword list (item 4b), closing the SES-coverage gap. Re-validated on real data in both native and Docker (v3.10.10): SocioeconomicStatus maps at 81% confidence (domain tier, p=7.17e-116, 10 groups, min size 2,536) and is now the most heavily-weighted significant feature in this dataset. SVM enforcement now runs on this dataset (previously skipped) and was correctly rejected by the gate: composite worsened (0.0186→0.0440) plus a new disadvantaged-group reversal on Region; the rebalanced stage was used as the final result. SocioeconomicStatus and DisabilityStatus each showed a technical group-rate reversal after rebalancing, but both were correctly held as non-actionable since the resulting gaps fell within near-parity (2.0pp and 0.0pp, both below their own disparity thresholds).

3.8 Business: bank-full

Target column	y
Records	45,211 → 45,211 (100% retention)
Bias score (native, v3.10.1)	0.0664 → 0.0776, rebalancing made the composite WORSE
Bias score (Docker, v3.10.5)	0.0169, SUPERSEDES the native record; correctly stays below mitigation threshold, no rebalancing attempted
SVM stage	Native run: correctly rejected a collapsed-constant-prediction result (first real-data confirmation of this Phase 3.5 safeguard)

Finding: Only Age is mapped in this domain; Business's ontology has no MaritalStatus category (open item 1, unaffected by this validation).

Finding: The native result is the clearest original example of the Age-bucketing open item: the worst-disparity age group (68, n=36) sat below the 50-sample floor and was structurally excluded, causing rebalancing to actually worsen the composite. The Docker log shows the bucketing fix firing cleanly: 77 raw values → 4 groups (min group size 10,279, comfortably above the floor), the fourth real-data cross-domain confirmation of the bucketing fix, after Governance, German Credit, and Heart Disease.

Finding: With clean bucketed groups, Docker's composite (0.0169, still 1 significant feature, Age, $p=5.13e-36$) correctly stays below the 0.05 mitigation threshold, so no rebalancing is attempted at all, eliminating any risk of the old regression recurring. Open item 2 (raw-per-year Age groups) is now effectively resolved for Business.

3.9 Governance: ACS Public Coverage (Washington, 2018)

Target column	PUBCOV
Records	24,312 → 24,377 (100.3% retention)
Bias score (native, v3.10.5, re-run #4)	0.2036 → 0.1310 (+35.7%)
Bias score (Docker, v3.10.5)	0.20363 → 0.13097 (+35.7%), clean parity, matches native exactly
SVM stage	Applied, REJECTED (classifier collapsed to a single constant prediction across all rows; introduced 5 new reversals, Age, Gender, DisabilityStatus, MigrationStatus, Ethnicity)
Rebalance gate	Accepted, with fuse exclusions (Ethnicity, Age)

Feature	Column used	Weight	p-value	Disparity: before → after
Ethnicity	race	0.25	<0.001	38.5pp → 34.0pp (excluded by fuse, reverted)
Gender	sex	0.20	<0.001	7.8pp → 6.0pp
SocioeconomicStatus	income	0.15	<0.001	23.8pp → 6.1pp (4 quantile buckets, min group size 5,805)
DisabilityStatus	disability	0.10	<0.001	43.8pp → 2.7pp
MigrationStatus	citizenship	0.10	<0.001	14.3pp → 8.8pp
Age	age	0.05	<0.001	27.4pp → 24.2pp (excluded by fuse, reverted)

Finding: Income was cleanly bucketed into 4 quantile ranges (min group size 5,805), including a legitimate '-7,400-0.00' negative-income bucket reflecting ACS's real business/farm-loss income convention, not a stray singleton artifact. This matches the native record exactly and is the primary real-data motivation behind the v3.10.5 bucketing fix.

Finding: The rebalance-stage fuse excludes exactly 2 features (Ethnicity, Age) after their rebalancing reversed the disadvantaged group, down from 3 exclusions in earlier pre-fix attempts on this same dataset, and matching the final native-venv confirmed state exactly in Docker.

Finding: SVM enforcement was correctly rejected for classifier collapse plus 5 new reversals; final predictions correctly fell back to the rebalanced stage rather than the collapsed SVM output.

Caveat, open item 9 (unresolved): The report's "Why this recommendation" section still lists all 6 rejected-SVM-stage reversals (Age, Gender, DisabilityStatus, MigrationStatus, SocioeconomicStatus, Ethnicity) as though actionable, even though the final delivered result uses only the rebalanced stage and none of the SVM-stage reversals apply to it. This wording gap is confirmed present and unchanged in the Docker re-run.

Caveat, open item 10 (unresolved): Fuse-excluded/reverted features can still collaterally drift as a side effect of other features' resampling, with no check currently catching this. Confirmed on real data: Age, despite being excluded and reverted to its original values, still shows disparity worsening (0.1717→0.1883, a -9.6% "improvement"); Ethnicity's worst group also shifted (6.0→5.0) purely from other features' resampling. Both are reconfirmed present, not worse, in the Docker re-run.

4. Pipeline Fixes and Findings from This Validation Round

Each fix below was root-caused to an exact code path, fixed, and verified against the real dataset that surfaced it, per the SOP's bug-handling protocol. The seventh entry is a root-caused finding, not a fix; it predates Phase 5 and is documented here for completeness and transparency rather than as a defect introduced during this round.

4.1 v3.10.1, Target-column auto-detection removed entirely

Found on: Hiring/OpenIntro_resume.csv, auto-detection skipped the true outcome received_callback (20% confidence, no keyword match) and fell back to job_fed_contractor purely because it appeared earlier in column order among 20 tied binary candidates.

Root cause: a structural failure, not a keyword-coverage gap; binary-vs-not carries zero separating power once many columns are binary. Cross-checked against AIF360/Fairlearn/Aequitas: none attempt target-column auto-detection at all, since "what does this dataset measure" is a real-world-purpose question only the dataset's owner can answer.

Fix: target_column is now required in process_dataset (clear ValueError citing precedent), CLI prompt no longer accepts blank Enter, HTML form field changed Optional→Required with disclaimer plus best-effort unranked column preview.

Verified: Hiring/OpenIntro_resume.csv re-confirmed on clean v3.10.1 provenance, bit-for-bit identical to the pre-fix workaround numbers.

4.2 v3.10.1, Delimiter auto-detection added

Found on: Business/bank-full.csv (semicolon-delimited), silently loaded as one column containing each entire row as a string, since every pd.read_csv() call assumed comma.

Fix: a shared smart_read_csv() helper (csv.Sniffer against comma/semicolon/tab/pipe, defaulting to comma if inconclusive) swapped into all pipeline read sites.

Verified: 6 test scenarios including the exact bank-full.csv header, comma-file regression, genuine single-column file (no false positive), tab-delimited, sniffer-defeating case, and a full round trip.

4.3 v3.10.2, CamelCase/PascalCase tokenization fixed

Found on: Finance/Preprocessed_german_credit.csv, PascalCase column names (AgeInYears, PersonalStatusAndSex, ForeignWorker) mapped zero of 21 columns to any protected attribute on the first run.

Root cause: the word-boundary check only treats non-alphanumeric characters as boundaries, so PascalCase names with zero separators never gave legitimate matches a boundary to land on.

Fix: map_column now inserts a word-boundary at every camelCase/PascalCase case transition before matching.

Verified: 9-unit tests plus a full end-to-end pipeline run; purely additive, no previously correct mapping changed anywhere.

4.4 v3.10.3→v3.10.4, Finance SES keyword coverage + p-value-based column selection

Found on: Finance/Preprocessed_german_credit.csv, the highest-weighted SocioeconomicStatus feature was only catching InstallmentRatePctOfDisposableIncome (a loan-structuring ratio, not genuine wealth), while real SES columns (SavingsAccountBonds/Property/Housing/PresentEmploymentSince/Job) went unrecognized.

Fix: added savings/property/housing/employment/job keywords to Finance's Tier-2 contextual_patterns (scoped to Finance only), and changed _select_best_column_per_feature's ranking from confidence-only to (small-group-warning, p-value, confidence) so strongest actual evidence wins over naming confidence alone.

Verified: 4 scenarios including real German Credit numbers (Property now correctly wins with $p=2.858e-05$, previously the weakest of six candidates by naming confidence alone).

4.5 v3.10.5, Rebalance-stage fuse

Found on: Governance/ACS PublicCoverage, the disadvantaged group reversed for Ethnicity, Age, and SocioeconomicStatus after rebalancing; the existing rebalance_gate correctly flagged this, but the flagged, reversed result was still delivered as final.

Fix: any feature whose rebalancing produces an actionable reversal is now excluded and reverted to original values, with remaining features redone from scratch, replacing the old flag-only behavior.

Verified: confirmed working across four Governance re-runs plus cross-domain on German Credit, Heart Disease, and Governance's Docker re-run; excluded_features now correctly populated in audit_trail.json with feature/reason/disadvantaged_before/after.

4.6 v3.10.5, Continuous protected-attribute bucketing

Found on: Governance/ACS PublicCoverage, income as 859 raw near-continuous values produced an illegible chart and a nonsensical negative-dollar singleton "disadvantaged group" after SVM.

Fix: any column mapped to Age or SocioeconomicStatus that's numeric and would trip the excessive-cardinality skip now gets quantile-bucketed (via `_quantile_bucket_numeric_column()`, max 4 bins, min_bin_size=50) before validation or rebalancing.

Verified: confirmed firing correctly and improving the validated result on five real datasets: Governance (income), German Credit (Age), Heart Disease (Age), Business (Age), Bar Pass (Age, both candidates). Datasets that already ship pre-bucketed age (studentInfo/OULAD) are correctly left untouched.

4.7 Investigated finding, closed as a platform characteristic, not a code defect, native-vs-Docker SVM result variance

Found on: Investigation: six candidate causes were systematically tested and ruled out, unseeded `fit()` (disproven: three fresh-process native re-runs produced bit-for-bit identical results), missing dependency pins (disproven: fairlearn/scikit-learn/numpy/scipy are exactly pinned and match installed versions), a changed CSV file (disproven: unmodified since before either run), Docker using a different requirements file (disproven: same shared pinned file), a Python version mismatch (disproven: 3.9.6 both sides), and Rosetta/QEMU emulation on Apple Silicon (disproven: Docker confirmed running natively as arm64).

Root cause: with every other variable ruled out and confirmed identical, the remaining explanation is inherent BLAS-backend variance, macOS's native numpy/scipy wheels link against Apple's Accelerate/vecLib, while the Linux container's wheels link against OpenBLAS. Same chip, same Python version, same package versions, but two different compiled numerical backends produce tiny floating-point differences that ExponentiatedGradient's iterative convergence amplifies into a small but real difference in the final result.

Confirmed via direct source inspection that this is byte-for-byte identical, and identically unseeded, in both v3.10.1 (the exact file that produced the native Hiring run) and v3.10.5, a standing characteristic of the `fair_reduction` method since its introduction, not a version regression or a Docker-specific defect.

Status (updated, v3.10.10 Docker sweep): this variance has NOT reproduced since the Phase 4 dependency-pinning work (Workstream D) pinned native and Docker to identical BLAS/scikit-learn/numpy/scipy versions. Hiring's Docker re-run under v3.10.10 matched native bit-for-bit (composite to all 15 decimal places, identical SVM predictions_changed count) — the exact case that originally diverged. The root-cause analysis above remains accurate for the pre-pinning setup; the finding is downgraded from "inherent/unfixable platform characteristic" to "observed under the pre-dependency-pinning setup, not reproduced since Workstream D." No further action needed unless it resurfaces.

5. Cross-Domain Patterns

The bucketing fix changed the validated outcome in five of nine datasets

Governance, German Credit, Heart Disease, Business, and Bar Pass all saw their Docker/post-v3.10.5 result diverge from an earlier native record purely because raw, near-continuous Age or SocioeconomicStatus columns were bucketed for the first time. In every case, the divergence is an improvement or correction of a previously understated result, not a regression, most strikingly Business, where the pre-fix native result showed rebalancing making the composite worse (0.0664→0.0776), a defect the bucketing fix directly eliminated.

SVM gate rejections span every distinct failure mode observed in Phase 5

Dataset	SVM outcome
Business (native)	Rejected, classifier collapsed to a constant prediction
Heart Disease	Rejected, plain composite-score regression, no collapse or reversal
Governance	Rejected, classifier collapse plus 5 new reversals
Hiring	Accepted, first Phase 5 dataset where the gate passed a genuinely-good SVM result

This spread, three distinct rejection reasons plus one genuine acceptance, confirms the gate's multiple criteria are each functioning independently, not just catching the same failure mode repeatedly.

Governance stands alone as the domain with unresolved, reconfirmed transparency gaps

Open items 9 and 10 (SVM-stage reversals cited as actionable in report wording; fuse-excluded features still collaterally drifting) were both found during Governance's native-venv saga and reconfirmed present and unchanged in the Docker re-run. No other domain in Phase 5 surfaced either issue, though both are architecturally possible anywhere the fuse or SVM rejection paths engage.

Result reproducibility is a newly surfaced, cross-cutting caveat

The fair_reduction SVM non-determinism (Section 4.7) applies to any dataset where SVM enforcement runs, not just Hiring; it was only visible in Hiring because that is the one Phase 5 dataset where the gate accepted the SVM result on both runs rather than rejecting it (rejection masks the underlying training variance, since the rejected result is discarded either way).

6. Recommendation

All nine dataset validations across six domains are complete, in both native and Docker environments, with root-caused explanations for every anomaly encountered. Six real, previously latent pipeline defects were found and fixed during Phase 5, each verified against the real data that surfaced it; a seventh finding (native-vs-Docker SVM result variance) was investigated and originally closed as inherent cross-platform BLAS-backend variance; a later v3.10.10 Docker sweep found the variance no longer reproduces, likely due to Workstream D's dependency pinning (see Section 4.7). A further round of open-item fixes was completed post-Phase-5 (pipeline now v3.10.10): eight items in the initial round (v3.10.6/3.10.7), plus three more surfaced during the full 9-dataset re-validation sweep (v3.10.8 features-key fix, v3.10.9 Hiring SES-keyword correction, v3.10.10 report-narrative rounding fix), detailed below.

Open items, resolved during Phase 5

- Item 2 (Age-as-raw-per-year-groups): resolved wherever the v3.10.5 bucketing fix fires, confirmed on Governance, German Credit, Heart Disease, Business, and Bar Pass.
- Items 3, 4, 6, 8, 9, 10 (v3.10.6) and Item 5 (v3.10.7): all closed in a post-Phase-5 hardening round ahead of handover, see the replacement list below in place of 'still open.,
Item 11: investigated; the original finding (platform-characteristic BLAS-backend variance) has not reproduced since Workstream D's dependency pinning, see Section 4.7 for the updated status.
- Item 1: closed as an accepted limitation, no 8th protected-attribute feature added, per policy decision.
- Item 12 (v3.10.8): a self.results features-key omission was found during v3.10.7 re-validation, silently broke mapping-tie winner-naming and the item-3 p-value transparency line in every delivered report.pdf since v3.10.4. Fixed and confirmed on HMDA real data.
- Item 13 (v3.10.9): Hiring's SocioeconomicStatus keyword list incorrectly matched years_experience as an SES proxy; removed per Hamid's decision. Re-confirmed on real data, reverted cleanly to the correct 2-feature baseline.
- Item 14 (v3.10.10): report-narrative rounding bug, the "Why this recommendation" text always showed the SVM-rejected score as unchanged ("from X to X") regardless of true magnitude. Fixed by storing the raw SVM score under a new key; confirmed on Heart Disease real data.

Open items: deferred, not BiasClean's to close

- Item 7: FDK's own equivalent auto-detection fix, out of scope for this report; deliberately left for Faria to action on FDK's own APIs, per Hamid's direction that FDK responsibility has already transferred to her.

Additional recommendations

- Pipeline is now v3.10.10. All fixes through v3.10.10 have been fully re-validated on real data across all 9 Phase 5 datasets, in both native venv and Docker environments, the full re-validation sweep found and fixed three further issues beyond the original eleven (items 12-14 above) and resolved item 11's open question (see Section 4.7). This handover is considered fully closed out.
- German Credit's Property column may now auto-approve at the default 0.80 threshold under item 5's fix (previously needed a manual 0.70 override), worth re-testing without the override.

Full machine-readable results (audit_trail.json) and formatted reports (report.pdf) for every dataset referenced in this document are retained alongside the corrected datasets they were generated from, per the SOP's own record-keeping requirements.